

CSA1717-ARTIFICIAL INTELLIGENCE
PRACTICAL LAB EXPERIMENTS
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MEDICAL DIAGNOSIS

Aim:

To implement a Prolog-based medical diagnosis system using symptoms and logical rules.

Objective:

To infer possible diseases based on symptoms provided by the user.

Algorithm

- 1. Start.**
- 2. Define symptoms.**
- 3. Define disease rules.**
- 4. Enter the patient's symptoms.**
- 5. Match symptoms with disease rules.**
- 6. Display the possible diagnosis.**
- 7. Stop.**

Flowchart

START

|

v

Enter Symptoms

|

v

Match Disease Rules

|

v

Required Symptoms Present?

/ \

Yes No

| |

Disease Check

Derived Next Rule

|

v

Display Diagnosis

|

v

STOP

Prolog Program

% Medical Diagnosis System

symptom(ravi, fever).

symptom(ravi, cough).

symptom(ravi, body_pain).

symptom(priya, fever).

symptom(priya, rash).

symptom(arun, cough).

symptom(arun, breathlessness).

diagnosis(Patient, flu) :-

symptom(Patient, fever),

symptom(Patient, cough).

diagnosis(Patient, measles) :-

symptom(Patient, fever),

symptom(Patient, rash).

diagnosis(Patient, pneumonia) :-

symptom(Patient, cough),

symptom(Patient, breathlessness).

Queries

?- diagnosis(ravi, Disease).

Output:

Disease = flu.

?- diagnosis(priya, Disease).

Output:

Disease = measles.

?- diagnosis(arun, Disease).

Output:

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Disease = flu.
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Result:

Thus, the medical diagnosis system successfully identifies possible diseases based on symptoms.